



Generalising linear relationships

Teacher resource

Question 1

A chef suggests the best cooking time for roast lamb is 25 minutes per kilogram plus 20 minutes.

If m is the weight of the lamb in (kilograms), the total cooking time in minutes is:

- ☐ $45m$
- ☐ $25m + 20$
- ☐ $20m + 25$
- ☐ $25(m + 20)$

Answer: B

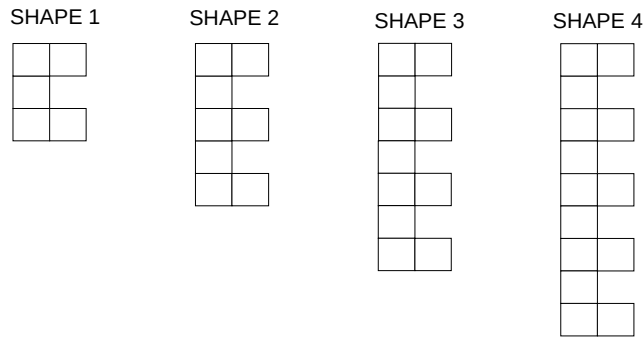
Skill: Students interpret an algebraic expression in a familiar context.

Additional questions

1. In the rule above, the additional 20 minutes is to allow for the meat to heat up before the actual cooking process begins. It then takes 25 minutes per kilogram for the piece of lamb to cook. How long will it take to cook a piece of lamb weighing three and a half kilograms?
2. Express the rule that shows the total cooking time C , in minutes, for a piece of lamb weighing p kilograms to cook as an algebraic equation.
3. Turkey takes 25 minutes plus 35 minutes per kilogram to cook. If it took a total of $4\frac{1}{2}$ hours to cook a turkey, what was the weight of the turkey?

Question 2

A pattern of shapes made of squares is shown below.



Choose the rule that can be used to find the number of squares in a shape if the Shape Number is known.

- ☐ Shape Number + 3
- ☐ $3 \times \text{Shape Number}$
- ☐ $3 \times \text{Shape Number} + 2$
- ☐ $5 \times \text{Shape Number}$

Answer: C

Skill: Students recognise and express a linear relationship in words and symbols

Background information

Students should develop a systematic approach to identifying rules to describe a pattern. When given a spatial pattern, students should be encouraged to represent the relevant numerical information in a table of values.

For example:

Tower number	1	2	3	4
Number of dots	5	8	11	14

Students may need to be prompted to look for the relationship between the pairs of numbers rather than just within the sequence of numbers in a row. They should understand that finding the relationship between the numbers within a row can help them to identify the relationship between the two variables. They also need to ensure the rule that they identify to describe the relationship between the variables works for all the pairs of numbers.

Additional questions

- Express this rule using variables.
- How many squares would be used in Shape Number 100 in this pattern?
- A shape in this pattern uses 737 squares. What is its Shape Number?

Question 3

Dora used the same rule to change a first number into a second number. She recorded the first and second numbers in the table below.

First number	0	4	1	121	h	13
Second number	1	9	3	243	81	27

What number is represented by h ?

Answer: 40

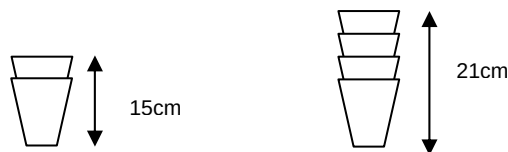
Skill: Students recognise and use a rule involving two operations

Additional questions

1. The first number is 0.5. What is the second number?
2. The second number is 15. What is the first number?
3. The second number is -12. What is the first number?

Question 4

Polystyrene cups are designed to stack inside each other.



A stack of two cups is 15 centimetres tall. A stack of four cups is 21 centimetres tall.

Which rule could be used to calculate the height of a stack of p cups?

- ☐ $3(p + 12)$
- ☐ $12p + 3$
- ☐ $3p + 12$
- ☐ $3p + 9$

Answer: D

Skill: Students generalise and write a linear rule.

Additional questions

1. What would be the height of a stack of 100 cups?
2. What is the smallest number of cups needed to make a stack that is at least one metre tall?

Student activity: Generalising linear relationships

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- ☐ $20m + 25$
- ☐ $25(m + 20)$

Question 3

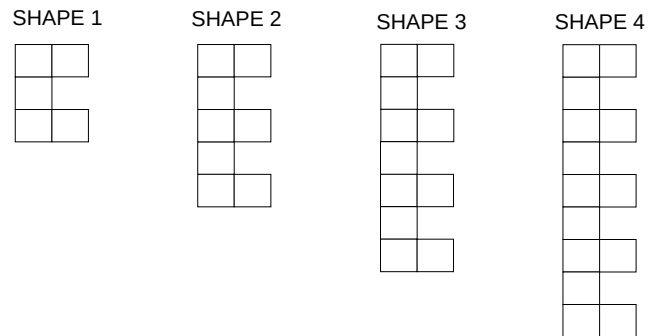
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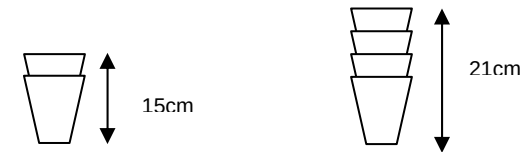


Choose the rule that can be used to find the number of squares in a shape if the Shape Number is known.

- ☐ Shape Number + 3
- ☐ $3 \times \text{Shape Number}$
- ☐ $3 \times \text{Shape Number} + 2$
- ☐ $5 \times \text{Shape Number}$

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